ASSIGNMENT -1

Task: Declare variables using var, let, and const to store a user's name, age, employment status, and a list of skills. Use typeof to check each variable's type and log them to the console.



ASSIGNMENT -2

Task: Use prompt() to capture a user's bill amount and tip percentage. Perform explicit type conversion (e.g., using Number()), compute the total using arithmetic and assignment operators, and log a formatted string.



ASSIGNMENT -3

Task: Create a script that evaluates a user's numerical test score using if...else statements, determines a letter grade using a switch statement, and uses a ternary operator to print whether they passed or failed.

main.js

Run ▶

```
let score = 85;
let grade;
if (score >= 90) grade = "A";
else if (score >= 80) grade = "B";
else if (score >= 70) grade = "C";
else if (score >= 60) grade = "D";
else grade = "F";
switch (grade) {
    case "A":
        grade = "A";
        break;
    case "B":
        grade = "B";
        break;
    case "C":
        grade = "C";
        break;
    case "D":
        grade = "D";
        break;
    default:
        grade = "F";
}
let status = score >= 40 ? "Passed" : "Failed";
console.log("Score: " + score);
console.log("Grade: " + grade);
console.log("Status: " + status);

```

Output

```
Score: 85
Grade: B
Status: Passed
```

ASSIGNMENT -4

Task: Write a for loop to print a multiplication table up to 5. Use a while loop with a break statement to find the first number greater than 10 that is divisible by 6, and a do...while loop that skips numbers using continue.

main.js

Run ▶

```
1  console.log("-- Multiplication Table of 5 --");
2  for (let i = 1; i <= 5; i++) {
3      console.log(`5 x ${i} = ${5 * i}`);
4  }
5  console.log("\n-- First number > 10 divisible by 6 --");
6  let num = 11;
7  while (true) {
8      if (num > 10 && num % 6 === 0) {
9          console.log("Found:", num);
10         break;
11     }
12     num++;
13 }
```

Output

```
-- Multiplication Table of 5 --
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25

-- First number > 10 divisible by 6 --
Found: 12
```

ASSIGNMENT -5

Task: Define a standard function and an arrow function that accept parameters (e.g., length and width) with default values to calculate the area of a rectangle. Demonstrate function and block scope limits.

```
function rectangleArea(length = 5, width = 4) {
    return length * width;
}
const arrowArea = (length = 5, width = 4) => {
    return length * width;
};
console.log(
    "Standard Function Area (Default 5x4):",
    rectangleArea()
);
console.log(
    "Arrow Function Area (Custom 10x3):",
    arrowArea(10, 3)
);
{
    let blockVar = 10;
}
try {
    console.log(blockVar);
} catch (error) {
    console.log("Scope Test Error:", error.toString());
}
```

Console 3

```
"Running fiddle"
"Standard Function Area (Default 5x4):", 20
"Arrow Function Area (Custom 10x3):", 30
"Scope Test Error:", "ReferenceError: blockVar is not defined"
>_
```

ASSIGNMENT -6

Task: Create an array of product objects (with name and price properties). Use forEach() to print each product, filter() to find items under a certain price, and map() to create an array of formatted name-price strings.

main.js

Run ▶

```
const products = [
    { name: "Laptop", price: 800 },
    { name: "Mouse", price: 25 },
    { name: "Keyboard", price: 45 }
];
console.log("-- All Products (forEach) --");
products.forEach(function(product) {
    console.log(`Item: ${product.name} - Price: $${product.price}`);
});
console.log("\n-- Filtered Products (Price < $50) --");
const filteredProducts = products.filter(function(product) {
    return product.price < 50;
});
console.log(filteredProducts);
console.log("\n-- Mapped Product Tags (map) --");
const productTags = products.map(function(product) {
    return `${product.name.toUpperCase()} costs $${product.price}`;
});
console.log(productTags);
```

Output

```
-- All Products (forEach) --
Item: Laptop - Price: $800
Item: Mouse - Price: $25
Item: Keyboard - Price: $45

-- Filtered Products (Price < $50) --
[ { name: 'Mouse', price: 25 }, { name: 'Keyboard', price: 45 } ]

-- Mapped Product Tags (map) --
[ 'LAPTOP costs $800', 'MOUSE costs $25', 'KEYBOARD costs $45' ]
```

CAPSTONE MINI PROJECT

Project Overview: Students will build an integrated console-based application that combines all core JavaScript concepts into a single comprehensive program.

```
console.log("--- Initializing Inventory System ---");
const inventory = [
    { name: "Laptop", price: 800 },
    { name: "Mouse", price: 25 }
];
const productName = prompt("Enter product name:");
const priceInput = prompt("Enter product price:");
const productPrice = Number(priceInput);
if (
    productName === null ||
    productName.trim() === "" ||
    priceInput === null ||
    priceInput.trim() === "" ||
    !Number.isFinite(productPrice) ||
    productPrice ≤ 0
) {
    console.log("Invalid input. Product not added.");
} else {
    const name = productName.trim();
    inventory.push({
        name: name,
        price: productPrice
    });
```

```
Console 11 0 0 0
"Running fiddle"
"--- Initializing Inventory System ---"
"[Prompt executed: User adds \"Monitor\" at \"$150\"]"
"
--- Processing Inventory Roster (forEach) ---"
"* Item: Laptop ($800) → Category: Premium"
"* Item: Mouse ($25) → Category: Budget"
"* Item: Monitor ($150) → Category: Standard"
"
--- Financial Analytics ---"
>_
```

JavaScript

```
    console.log(
        `[Prompt executed: User adds "${name}" at "$${productPrice}"]`
    );
    function getCategory(price) {
        if (price < 50) {
            return "Budget";
        } else if (price < 200) {
            return "Standard";
        } else {
            return "Premium";
        }
    }
    function getBadge(category) {
        switch (category) {
            case "Budget":
                return "Budget";
            case "Standard":
                return "Standard";
            case "Premium":
                return "Premium";
            default:
                return "Unknown";
        }
```

```
Console 11 0 0 0 Clear console Minimiz
"* Item: Laptop ($800) → Category: Premium"
"* Item: Mouse ($25) → Category: Budget"
"* Item: Monitor ($150) → Category: Standard"
"
--- Financial Analytics ---"
"Total Inventory Value: $975"
"Filtered Affordable Items (Under $200): [\"Mouse\",\"Monitor\"]"
"
Formatted Report List: [\"LAPTOP: $800\",\"MOUSE: $25\",\"MONITOR: $150\"]"
"Laptop price after discount and tax: $756.00"
>_
```

```
    console.log("\n--- Processing Inventory Roster (forEach) ---");
    inventory.forEach(item => {
        console.log(
            `* Item: ${item.name} ($${item.price}) → Category: ${getBadge(getCategory(item.price))}`
        );
    });
    let monitorFound = false;
    for (let i = 0; i < inventory.length; i++) {
        if (inventory[i].name.toLowerCase() === "monitor") {
            monitorFound = true;
            break;
        }
    }
    const calculateTotal = (items) => {
        let total = 0;
        for (const item of items) {
            if (item.price ≤ 0) {
                continue;
            }
            total += item.price;
        }
        return total;
    };
```

```
    const calculateTotal = (items) => {
    };
    function calculateFinalPrice(price, discount = 0, tax = 0) {
        const discountedPrice = price - (price * discount / 100);
        return discountedPrice + (discountedPrice * tax / 100);
    }
    console.log("\n--- Financial Analytics ---");
    console.log("Total Inventory Value: $" + calculateTotal(inventory));
    const affordableItems = inventory.filter(item => item.price < 200);
    console.log(
        "Filtered Affordable Items (Under $200): " +
        JSON.stringify(affordableItems.map(item => item.name))
    );
    const report = inventory.map(
        item => `${item.name.toUpperCase()}: $${item.price}`
    );
    console.log("\nFormatted Report List: " + JSON.stringify(report));
    console.log(
        "Laptop price after discount and tax: $" +
        calculateFinalPrice(800, 10, 5).toFixed(2)
    );
}
```

(property) price: number